Unit 2

Project Scheduling

2.1 Experiment 1: Project Evaluation and Review Technique (PERT) Analysis

2.1.1 Work Breakdown Structure (WBS) for Sanskrit NLP Project

The Work Breakdown Structure (WBS) is a hierarchical decomposition of the project into manageable activities [2]. The Sanskrit NLP Framework project is divided into nine major activities:

Table 2.1: Work Breakdown Structure with Time Estimates (in days)

Task ID	Activity Description	Optimistic (O)	Pessimistic (P)	Most Likely (M)
A	Requirement gathering & stakeholder sign-off	10	20	14
B	Corpus collection, digitization & expert annotation	30	50	38
C	Panini rule formalization (Ashtadhyayi sutras)	40	70	52
D	Morphological analyzer development	35	60	45
E	Semantic parser development	30	55	40
F	Knowledge extraction module development	20	40	28
G	API development & documentation	15	30	20

H	Integration testing & accuracy validation	20	40	28
I	Deployment, user training & project closure	10	20	14

2.1.2 PERT Expected Duration Calculation

The PERT expected duration formula [2] is:

$$T_e = \frac{O + 4M + P}{6}$$

Where:

- O = Optimistic time estimate
- M = Most likely time estimate
- P = Pessimistic time estimate

Table 2.2: PERT Expected Duration Calculation

Activity	**Calculation**	**Expected Duration (days)**
A	$(10 + 4 \times 14 + 20)/6$	14.33
B	$(30 + 4 \times 38 + 50)/6$	38.67
C	$(40 + 4 \times 52 + 70)/6$	53.00
D	$(35 + 4 \times 45 + 60)/6$	45.83
E	$(30 + 4 \times 40 + 55)/6$	40.83
F	$(20 + 4 \times 28 + 40)/6$	28.67
G	$(15 + 4 \times 20 + 30)/6$	20.83
H	$(20 + 4 \times 28 + 40)/6$	28.67
I	$(10 + 4 \times 14 + 20)/6$	14.33

2.1.3 Variance and Standard Deviation

Variance formula:

$$\sigma^2 = \left(\frac{P - O}{6}\right)^2$$

Standard deviation:

$$\sigma = \sqrt{\sigma^2}$$

Table 2.3: Variance and Standard Deviation by Activity

Activity	**Calculation**	**Variance** (σ^2)	**Std Dev** (σ)
A	$((20-10)/6)^2$	2.78	1.67
B	$((50-30)/6)^2$	11.11	3.33
C	$((70-40)/6)^2$	25.00	5.00
D	$((60-35)/6)^2$	17.36	4.17
E	$((55-30)/6)^2$	17.36	4.17
F	$((40-20)/6)^2$	11.11	3.33
G	$((30-15)/6)^2$	6.25	2.50
H	$((40-20)/6)^2$	11.11	3.33
I	$((20-10)/6)^2$	2.78	1.67

2.1.4 Probability Analysis

Total variance along critical path (identified in CPM analysis):

$$\sigma^2_{total} = \sigma^2_A + \sigma^2_C + \sigma^2_D + \sigma^2_E + \sigma^2_G + \sigma^2_H + \sigma^2_I$$

$$\sigma^2_{total} = 2.78 + 25.00 + 17.36 + 17.36 + 6.25 + 11.11 + 2.78 = 82.64$$

$$\sigma_{total} = \sqrt{82.64} = 9.09 \text{ days}$$

Probability of completing the project within 230 days:

$$Z = \frac{230 - 218}{9.09} = 1.32$$

From standard normal distribution table: $P(Z \leq 1.32) \approx 0.9066$ or 90.66%

2.2 Experiment 2: Critical Path Method (CPM) Analysis

2.2.1 Precedence Relationships

Table 2.4: Activity Precedence Relationships

Activity	**Predecessor(s)**	**Rationale**
A	-	Starting activity – requirements must be gathered first
B	A	Corpus selection depends on requirements

C	A	Panini rule formalization depends on requirements
D	B, C	Morphological analyzer requires both corpus and rules
E	D	Semantic parser depends on morphological analyzer
F	D	Knowledge extraction depends on morphological analyzer
G	E, F	API requires both semantic parser and knowledge extraction
H	G	Testing requires completed API
I	H	Deployment requires successful testing

2.2.2 Forward Pass (Earliest Start & Earliest Finish)

Assuming project start at day 0:

Table 2.5: Forward Pass Calculation

Activity	**Earliest Start (ES)**	**Earliest Finish (EF)**
A	0	14.33
B	14.33	53.00
C	14.33	67.33
D	67.33	113.16
E	113.16	154.00
F	113.16	141.83
G	154.00	174.83
H	174.83	203.50
I	203.50	217.83

2.2.3 Backward Pass (Latest Start & Latest Finish)

Table 2.6: Backward Pass and Slack Calculation

Activity	**Latest Start (LS)**	**Latest Finish (LF)**	**Slack (LS - ES)**
A	0	14.33	0
B	28.67	67.34	14.34
C	14.34	67.34	0.01

D	67.34	113.17	0.01
E	113.17	154.00	0.01
F	125.33	154.00	12.17
G	154.00	174.83	0
H	174.83	203.50	0
I	203.50	217.83	0

2.2.4 Critical Path Identification

Activities with zero slack constitute the critical path:

$$\mathbf{A} \rightarrow \mathbf{C} \rightarrow \mathbf{D} \rightarrow \mathbf{E} \rightarrow \mathbf{G} \rightarrow \mathbf{H} \rightarrow \mathbf{I}$$

Total Project Duration: 217.83 days (approximately 218 days or 10.5 months)

Note: Activity B (corpus annotation) has 14.34 days of slack and can be delayed without affecting the project completion date. Activity F (knowledge extraction) has 12.17 days of slack.

2.2.5 Gantt Chart Representation

Table 2.7: Gantt Chart Summary (Activities by Time Period)

Activity	**Duration**	**Timeline (Days)**
A	14 days	Days 0-14
B	39 days	Days 14-53
C	53 days	Days 14-67
D	46 days	Days 67-113
E	41 days	Days 113-154
F	29 days	Days 113-142
G	21 days	Days 154-175
H	29 days	Days 175-204
I	14 days	Days 204-218

2.3 Experiment 3: Project Resource Levelling Techniques

2.3.1 Project Team Formation and Structure

The project team consists of the following roles:

Table 2.8: Project Team Structure

Role	Number	Key Responsibilities
Project Manager	1	Overall planning, execution, stakeholder communication
Lead Sanskrit Linguist	1	Grammar validation, rule accuracy
Sanskrit Linguists	2	Annotation, rule formalization
Lead NLP Engineer	1	Architecture design, technical decisions
NLP Engineers	3	Development, testing, documentation
Data Scientist	1	Corpus processing, model training
QA Engineer	1	Testing, quality assurance

2.3.2 Selection of Project Manager

The Project Manager was selected based on the following criteria:

- Minimum 5 years of experience in software project management
- Prior experience in NLP or AI projects
- PMP certification preferred
- Excellent stakeholder management skills
- Experience with academic and government-funded projects

2.3.3 Roles and Responsibilities of Project Manager

Table 2.9: Project Manager Roles and Responsibilities

Area	Responsibilities
Planning	Develop project plan, WBS, schedule, and budget
Execution	Coordinate team activities, manage resources
Monitoring	Track progress, EVM analysis, risk monitoring
Communication	Stakeholder reporting, steering committee meetings
Quality	Ensure deliverables meet acceptance criteria
Closure	Final reporting, knowledge transfer, lessons learned

2.3.4 Project Execution Strategy

Project execution follows a phased approach:

1. **Phase 1 (Days 0-14)**: Requirement finalization and team onboarding
2. **Phase 2 (Days 14-67)**: Corpus collection and Panini rule formalization
3. **Phase 3 (Days 67-154)**: Core NLP module development
4. **Phase 4 (Days 154-204)**: API development and testing
5. **Phase 5 (Days 204-218)**: Deployment and closure

2.3.5 Resource Requirements by Activity

The project requires two types of critical resources: NLP Engineers and Sanskrit Linguists.

Table 2.10: Resource Requirements by Activity

Activity	Duration (days)	NLP Engineers	Linguists	Total Resources
A	14.33	1	2	3
B	38.67	2	3	5
C	53.00	2	4	6
D	45.83	3	2	5
E	40.83	3	2	5
F	28.67	2	1	3
G	20.83	2	0	2
H	28.67	2	2	4
I	14.33	1	1	2

2.3.6 Resource Availability Constraints

- **NLP Engineers available**: 4 (maximum)
- **Sanskrit Linguists available**: 4 (maximum – constraint based on expert availability)
- **Current peak NLP requirement**: 5 engineers (during activities D and E overlapping)
- **Current peak Linguist requirement**: 6 linguists (during activity C)

2.3.7 Resource Profile Before Levelling

Table 2.11: Resource Profile Before Levelling

Time Period (days)	**Activities**	**NLP Required**	**Linguists Required**	**Constraint Violation**
0 – 14	A	1	2	None
14 – 53	B, C	4 (2+2)	7 (3+4)	Linguists (7 > 4)
53 – 67	C	2	4	None (at limit)
67 – 113	D, E, F	8 (3+3+2)	5 (2+2+1)	NLP (8 > 4); Linguists (5 > 4)
113 – 141	E, F	5 (3+2)	3 (2+1)	NLP (5 > 4)
141 – 154	E	3	2	None
154 – 175	G	2	0	None
175 – 203	H	2	2	None
203 – 217	I	1	1	None

2.3.8 Resource Levelling Strategy

To resolve resource over-allocation, the following strategies are applied:

Strategy 1 – Shift Non-Critical Activities:

- Activity B (slack = 14.34 days) – shift start by 10 days to reduce overlap with C
- Activity F (slack = 12.17 days) – shift start by 15 days to reduce overlap with D and E

Strategy 2 – Split Critical Activities:

- Activity C (Panini rule formalization) – split into two sub-phases:
  - C1: 30 days with 4 linguists
  - C2: 23 days with 2 linguists

Strategy 3 – Add Additional Resource (if budget permits):

- Hire 1 additional linguist on contract (INR 1.5 Lakhs for 3 months)
- Hire 1 additional NLP engineer (INR 2 Lakhs for 4 months)

2.3.9 Resource Profile After Levelling

Table 2.12: Resource Profile After Levelling

Time Period (days)	NLP Required	Linguists Required	Status
0 – 14	1	2	OK
14 – 44	4	4	OK (after shifting B and splitting C)
44 – 67	3	3	OK
67 – 98	4	3	OK
98 – 113	5	3	OK (NLP at limit)
113 – 141	4	2	OK
141 – 154	3	2	OK
154 – 175	2	0	OK
175 – 203	2	2	OK
203 – 232	1	1	OK

2.3.10 Impact of Resource Levelling

Table 2.13: Resource Levelling Impact Summary

Parameter	Before Levelling	After Levelling
Peak NLP Engineers Required	8	5
Peak Linguists Required	7	4
Resource Constraint Violations	3 periods	0 periods
Project Duration	218 days	232 days
Schedule Increase	-	14 days (6.4%)
Additional Cost	-	INR 3.5 Lakhs (if hiring)

Recommendation: Accept the 14-day schedule increase instead of hiring additional resources, as the project has no fixed external deadline. This saves INR 3.5 Lakhs.

2.4 Experiment 4: Project Crashing Techniques

2.4.1 Crash Analysis – Rationale

Project crashing is considered only if:

1. A fixed deadline of 200 days is imposed by the sponsor
2. The 14-day increase from resource levelling is unacceptable
3. Budget contingency (INR 4–5 Lakhs) can absorb crash costs

2.4.2 Activity Crash Options

Table 2.14: Activity Crash Options

Activity	**Normal (days)**	**Crash (days)**	**Crash Cost per Day (INR)**	**On Critical Path?**
C (Panini formalization)	53.00	40	25,000	Yes (critical)
D (Morphological analyzer)	45.83	35	30,000	Yes (critical)
E (Semantic parser)	40.83	32	28,000	Yes (critical)
G (API development)	20.83	16	22,000	Yes (critical)
H (Testing & validation)	28.67	22	18,000	Yes (critical)
B (Corpus annotation)	38.67	32	12,000	No (slack = 14 days)
F (Knowledge extraction)	28.67	24	15,000	No (slack = 12 days)

2.4.3 Least Cost Crashing – Step by Step

Initial Critical Path: A – C – D – E – G – H – I (Duration: 218 days)

Target: 200 days (crash requirement: 18 days)

Table 2.15: Least Cost Crashing Steps

Step	**Action**	**Days Crashed**	**Cumulative Crash**	**Cost (INR)**
1	Crash activity E (cheapest on critical path) – reduce from 41 to 36 days	5	5	$5 \times 28,000 = 1,40,000$

2	Crash activity D – reduce from 46 to 41 days	5	10	$5 \times 30,000 = 1,50,000$
3	Crash activity G – reduce from 21 to 18 days	3	13	$3 \times 22,000 = 66,000$
4	Crash activity H – reduce from 29 to 24 days	5	18	$5 \times 18,000 = 90,000$
Total Crash Cost				**4,46,000**

2.4.4 Crash Impact on Non-Critical Paths

After crashing critical activities by 18 days, the project duration becomes 200 days:

Table 2.16: Non-Critical Path Impact

Path	**Original Duration**	**Duration After Crashing**
A – B – D – E – G – H – I	201 days (via B)	183 days (B unchanged)
A – C – D – F – G – H – I	193 days	175 days

All non-critical paths remain below 200 days. No secondary critical path emerges.

2.4.5 Crash Summary

Table 2.17: Crash Summary

Parameter	**Normal Schedule**	**Crashed Schedule**
Project Duration	218 days	200 days
Total Crash Cost	-	INR 4,46,000
Schedule Reduction	-	18 days
Average Cost per Day Saved	-	INR 24,778
Budget Contingency Available	INR 4–5 Lakhs	INR 4–5 Lakhs
Crash Within Contingency?	-	Yes

2.4.6 Crash Recommendation

- **If deadline is 218 days (baseline)**: No crashing required – proceed with normal schedule

- **If deadline is 200 days (sponsor requirement)**: Execute crash plan (INR 4,46,000) – fully within contingency budget
- **If deadline is 210 days (intermediate)**: Crash only steps 1 and 2 (INR 2,90,000) for 10 days saving

2.4.7 Time-Cost Trade-off Curve Summary

Table 2.18: Time-Cost Trade-off Summary

Duration (days)	**Total Crash Cost (INR)**	**Marginal Cost (INR/day)**	**Feasibility**
218	0	-	Baseline
210	2,90,000	36,250	Recommended for moderate compression
205	3,56,000	13,200	Possible
200	4,46,000	18,000	Maximum recommended
195	5,80,000 (estimated)	26,800	Not recommended (diminishing returns)

2.4.8 Conclusion for Unit 2

The PERT analysis provides a 90.66% probability of completing the project within 230 days [2]. The CPM analysis identifies the critical path A–C–D–E–G–H–I with a baseline duration of 218 days. The project team structure includes 10 members with clearly defined roles and responsibilities. Resource levelling resolves linguist and engineer over-allocation by shifting non-critical activities, increasing duration by 14 days. Crashing can reduce the schedule to 200 days at an additional cost of INR 4,46,000, which is within the contingency budget. The recommended approach is to proceed with the baseline schedule unless a fixed deadline requires crashing [8].